

Driver Drowsiness Detection using Digital Image Processing

Team Members

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Problem Statement

Driver fatigue is a leading contributor to road accidents, yet most vehicles lack any onboard mechanism to detect drowsiness before it causes a crash. This project builds a vision-based driver drowsiness detection system that classifies eye state (open/closed) from image data and, at inference time, tracks eyelid-closure behaviour over live video to raise a timely alert. The system is trained on cropped infrared eye images rather than full faces, so the methodology must bridge from a static, cropped-eye training source to temporal, whole-face live inference.

Existing Solutions & Research Gap: Widely used open-source pipelines (e.g. the PyImageSearch dlib+EAR tutorial and its many GitHub derivatives) rely on a single, manually tuned Eye Aspect Ratio (EAR) threshold applied uniformly to all users. This fails across individuals with different eye shapes, users wearing eyeglasses (glare/occlusion), and low-light or infrared night-driving conditions. Deep-learning classifiers (CNN, transformer) raise raw accuracy but are typically evaluated on clean, well-lit datasets and do not address these preprocessing failure modes. Commercial systems (Bosch, Volvo, Seeing Machines) achieve robustness with expensive dedicated hardware. This project targets the gap with a DIP-heavy enhancement/segmentation front-end and adaptive thresholding instead of a fixed EAR cutoff.

Objectives

- Curate and characterize the MRL Eye Dataset for eye-state classification.
- Design an enhancement front-end (CLAHE, gamma, glare suppression) for low-light/IR crops.
- Segment sclera/iris/pupil regions and clean masks via morphological operations.
- Apply DFT/DCT transforms for blur rejection and Hough-circle iris fitting.
- Implement adaptive-threshold EAR/MAR and PERCLOS at inference time on live video.
- Quantify JPEG/DCT compression trade-offs for in-vehicle frame transmission.

Proposed Methodology

The pipeline separates a **training stage** on the MRL Eye Dataset from an **inference stage** on live webcam/video. In training, cropped infrared eye images are size-normalized, enhanced with CLAHE and gamma correction, and segmented via adaptive/Otsu thresholding to isolate sclera, iris, and pupil; morphological opening/closing and hole-filling clean the resulting masks before a classifier is trained to predict eye state (open/closed). DFT/DCT-based sharpness checks filter blurred training frames, and Hough circle transforms fit the iris boundary for auxiliary shape features. At inference time, a Haar-cascade or MediaPipe face/eye localizer extracts eye and mouth regions from live frames; Eye Aspect Ratio (EAR) and Mouth Aspect Ratio (MAR) are computed on these localized landmarks — never on MRL crops directly, since MRL contains no facial landmarks — and combined with an adaptive (per-user-calibrated) threshold rather than one fixed globally. PERCLOS is accumulated over a rolling window of a video sequence (validated on a self-recorded clip) to give a temporally robust fatigue signal. Finally, a JPEG-quality/DCT sweep quantifies the accuracy-vs-bitrate trade-off relevant to transmitting frames from an in-vehicle camera to a fleet server.

DIP Topic	Pipeline Stage
Image enhancement	CLAHE + gamma correction on low-light IR eye crops; reflection suppression for eyeglasses
Segmentation	Sclera/iris/pupil isolation via adaptive + Otsu thresholding
Morphological ops	Opening/closing to clean eye mask; hole-filling before area measurement
Transforms	DFT/DCT for blur & motion detection; Hough circle transform for iris fitting
Image compression	DCT/JPEG quality sweep quantifying accuracy-vs-bitrate for in-vehicle transmission

Dataset Source

MRL Eye Dataset, VSB — Technical University of Ostrava (<https://mrl.cs.vsb.cz/data/eyedataset/>): 84,898 grayscale infrared eye-crop images from 37 subjects, annotated with eye state, gaze direction, glasses, reflections, and lighting condition. Images are pre-cropped eye patches of variable size (not full faces), so they serve as the

training/validation source for the eye-state classification stage only; EAR/MAR/PERCLOS are computed at inference time on live frames after face/eye localization, not on the dataset itself. A curated 40-image balanced sample (20 open/20 closed) is used for Week-1 exploratory analysis.

Expected Outcomes

A trained eye-state classifier evaluated on held-out MRL subjects; a documented DIP preprocessing pipeline (enhancement, segmentation, morphology, transforms) with before/after visualizations; a working live-inference demo computing adaptive EAR/MAR and PERCLOS on a self-recorded video with an audible/visual drowsiness alert; and a quantified JPEG compression-vs-accuracy curve. Collectively these deliverables demonstrate all five mandated DIP topics within one coherent, evaluated system.

Software Tools to be Used

Python 3.11, OpenCV, NumPy, scikit-image, Matplotlib, Pillow, imutils, Jupyter notebooks, Haar cascade / MediaPipe (face and eye localization, inference stage), Google Chrome (headless PDF rendering for reports), Git/GitHub (version control).